

Concurrent Collections and Parallel Streams



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Java Collections

Groups of related data items

Arrays have limitations

Java Collections Framework

Manipulate object references



Java Collections Framework

java.util.Collection

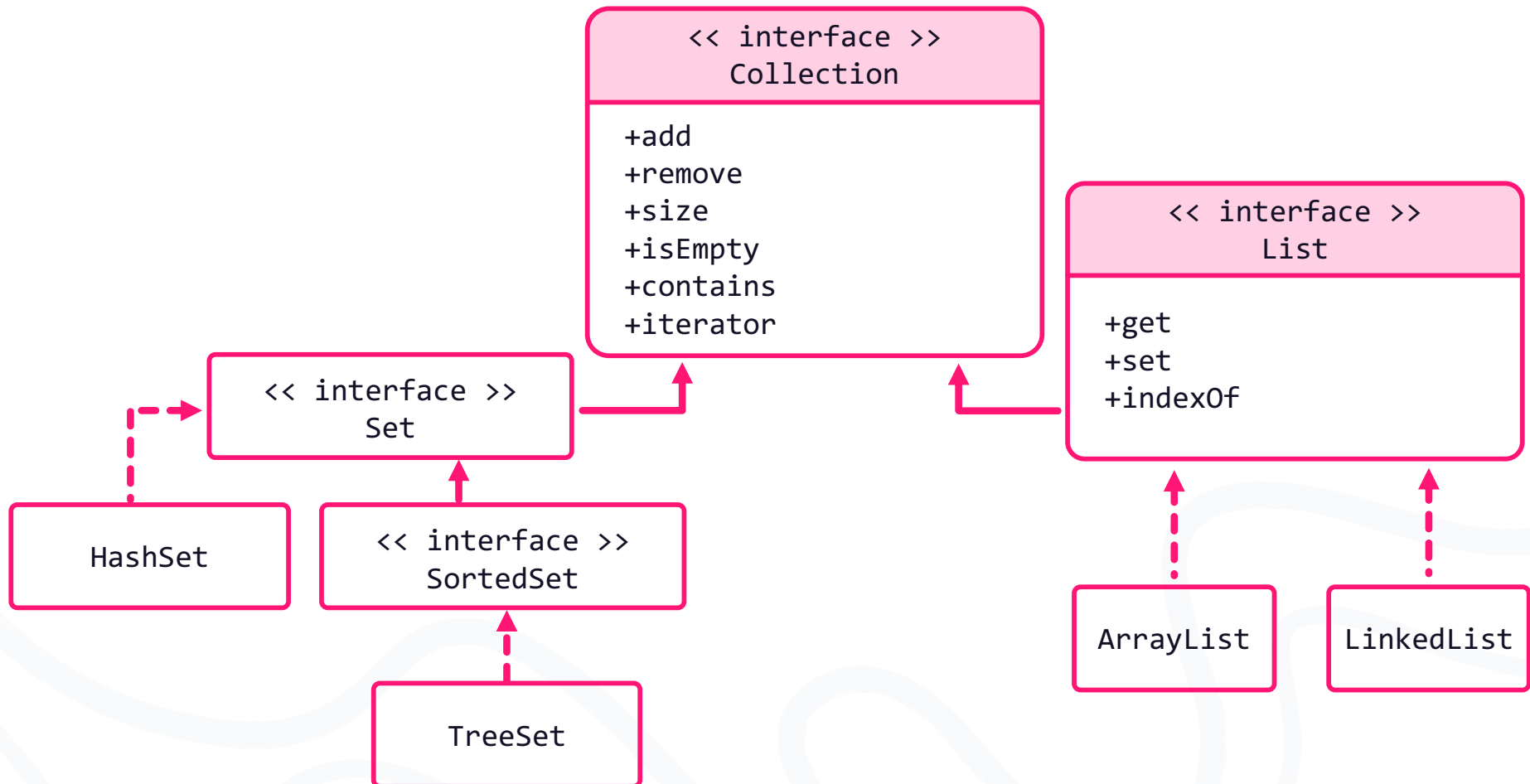
java.util.List

java.util.Set

java.util.Map



The Collections API



Synchronizing Collections

**Collections are not
synchronized**

**Methods are not
synchronized**

**Boosts single-
threaded programs**



Synchronizing Collections

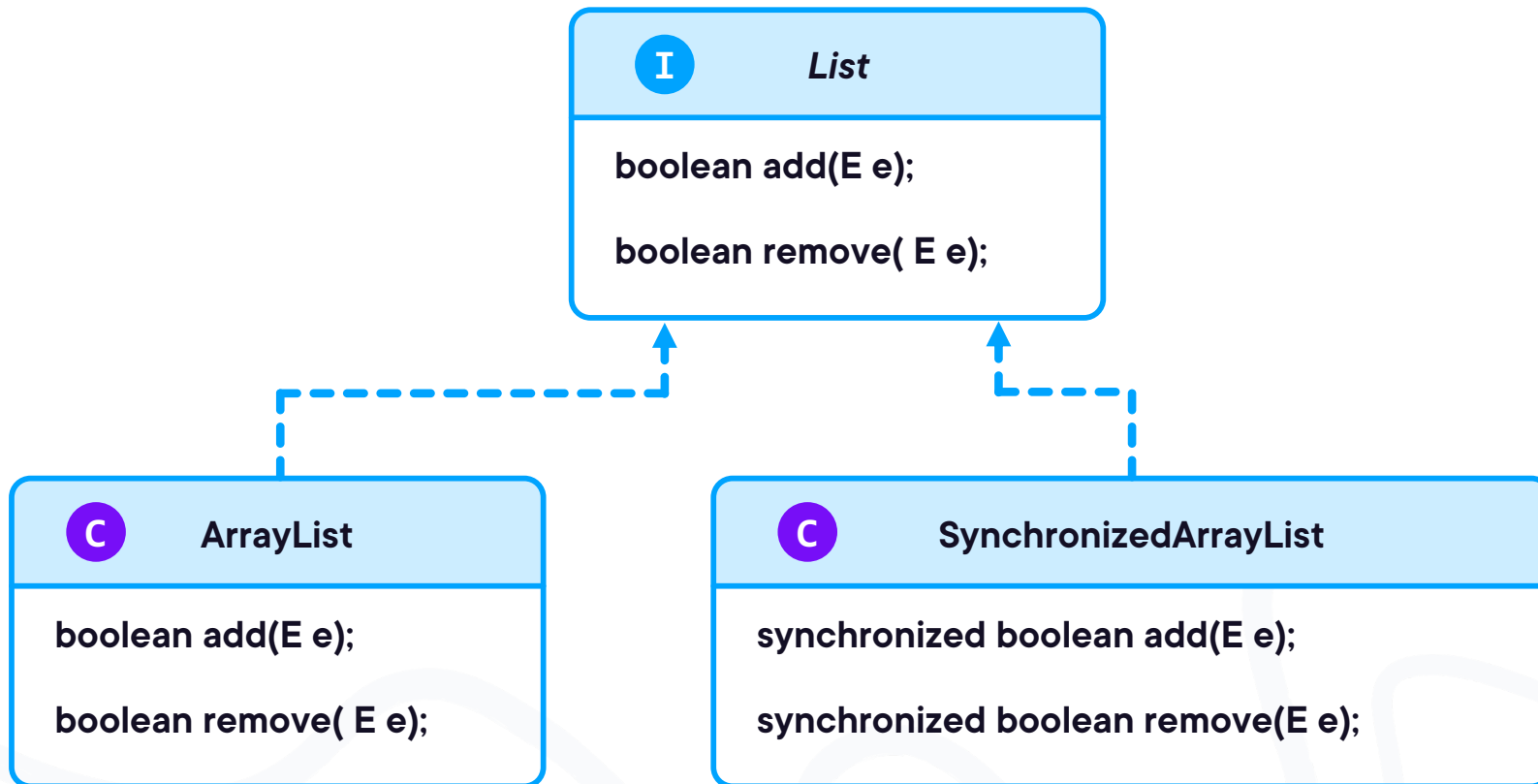
Synchronized access

**Collections.
synchronizedXYZ**

Proxied collection



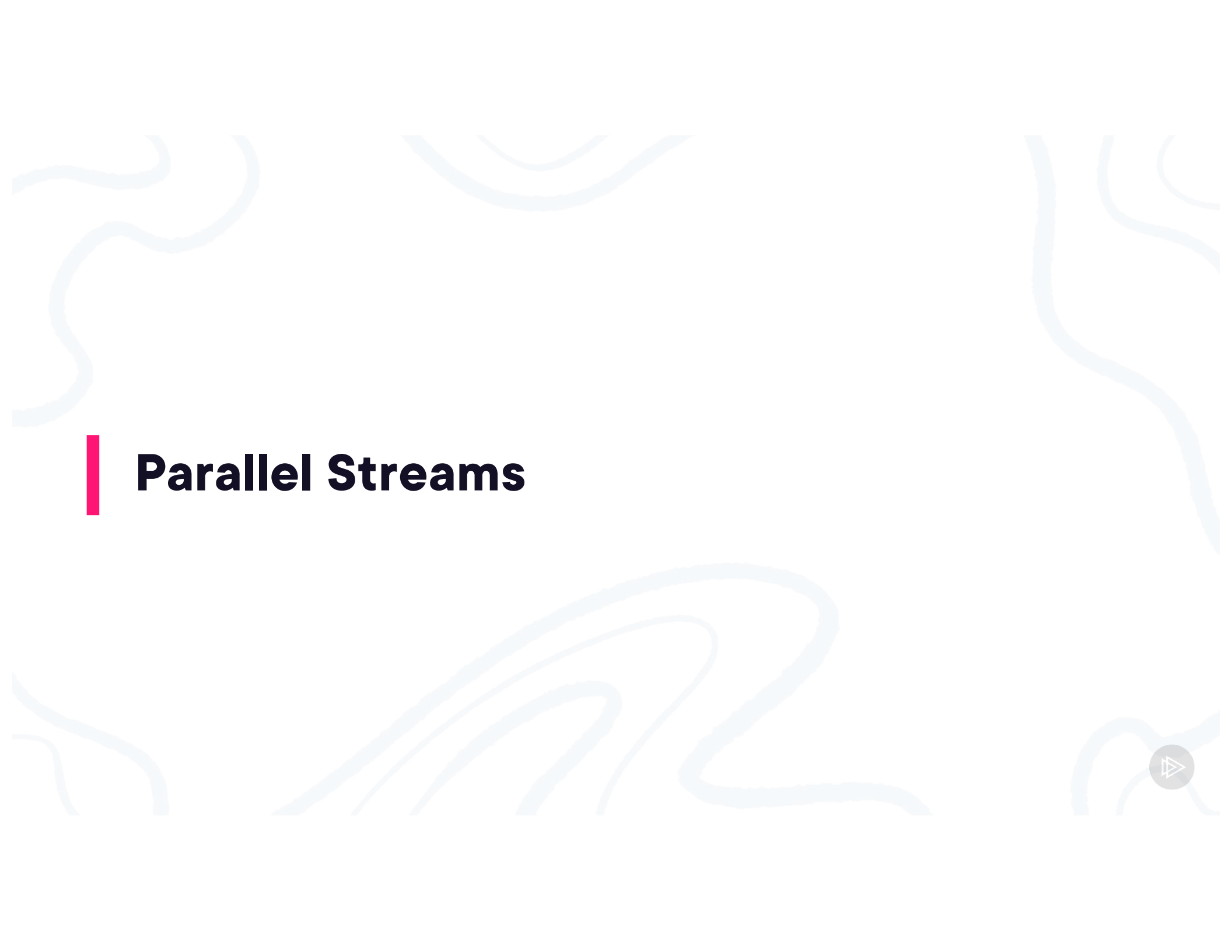
Synchronizing Collections





Demo: Synchronized Collections





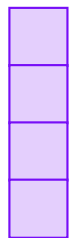
Parallel Streams



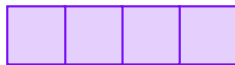
Introducing Streams

```
public interface Stream<T> extends BaseStream<T, Stream<T>> {  
    // ...  
}
```

Source



Stream



Operation



Stream



Terminal



Streams Features

**Can define
operations**

**Does not hold any
data**

**Does not change the
data it processes**

**Processed data in
« one pass »**

**Optimized from the
algorithm point of
view**



Typical Operations for Streams

Map

Filter

Reduce





Demo: Parallel Streams

